

Operating Systems Lab Assignment

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Course: B.Tech CSE Sec B

Assignment Title: Lab Environment Setup

Requirements/Software Used

- Ubuntu Linux (or WSL)
- Python 3.x
- Visual Studio Code
- **uv** (Python package manager)

Aim

To set up and configure the required environment for the Operating Systems Lab (Ubuntu Linux, WSL, VS Code, and Python) and verify each component is working correctly.

Approach/Algorithm

1. **Verification Script:** Wrote a Python script to automatically verify the environment.
2. **Python Check:** Used the built-in `sys.version` to verify the Python installation.
3. **OS Check:** Read from `/proc/version` to determine if the script is running under WSL, Ubuntu, or generic Linux.
4. **VS Code Check:** Used the `subprocess` module to run `code --version` to verify Visual Studio Code is installed and accessible in the PATH.

Output Screenshot

```
> cd /home/varun/web/os-assignment/2
uv run verify_env.py

Environment Verification:
- Python Version: 3.12.14
- OS Environment: Linux
- VS Code: Installed (Version: 1.140.0)

All required components checked.
```

Conclusion

The environment has been successfully set up and verified. The script properly detected the Linux/WSL environment, the Python version, and the VS Code installation, confirming readiness for future lab experiments.

GitHub Repository

<https://github.com/chauhan-varun/os-assignment>